

Python Radio 58: Make Your Own Semiconductors



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The Next Step in Building Our Radio

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Wikipedia

Before the invention of vacuum tubes, we received radio signals using semiconductors.

You couldn't just go to Amazon.com and buy a diode, however. You had to make one yourself. Eventually, companies started making "crystal detectors" and receivers, but you still had to play with the "cat's whisker" wire to find the right place on the crystal to get the best reception.

In the previous projects in this series, we built high-Q coils and homemade capacitors to tune our homemade radio. Now it is time to make our own diode detectors for AM reception.

But before we do, let's build some test equipment so we can tell if we have succeeded, and if so, how well.

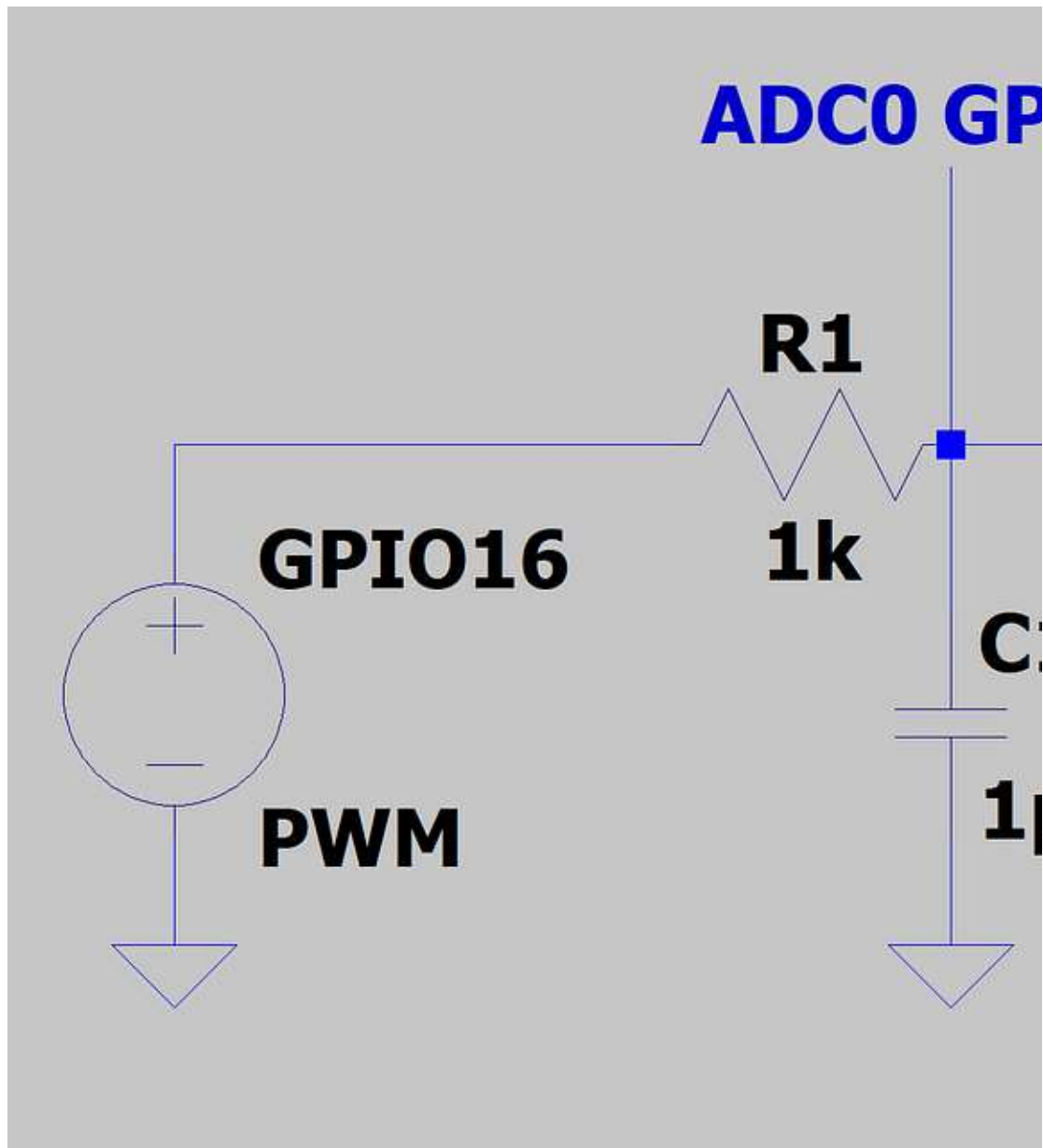
We'll continue using MicroPython on the RP2350 (Raspberry Pi Pico 2) microcontroller.

We want to ramp up the voltage from 0 to 3.3 volts and measure the current going through our diode. We'll do that by using our pulse width modulator followed by a low-pass filter to get a DC voltage. Since our analog-to-digital converter can only tell us about voltage, we will convert the current to a voltage using a 100-ohm shunt resistor, letting [Ohm's Law](#) work for us.

The Circuit

Our circuit looks like this:

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Resistor R1 and capacitor C1 form our low-pass filter. ADCo will record the voltage we are sending to the diode we are testing. ADC1 will record the voltage across the shunt resistor R2. It's pretty simple.

The Code

Here is the Python code:

```
import machine
import time
import math

# ===== USER CONFIGURATION
```

```

=====
# --- Hardware Pins ---
PWM_PIN_NUM = 16
ADC_V_APPLIED_PIN_NUM = 26 # ADC0
ADC_V_SHUNT_PIN_NUM = 27 # ADC1

# --- Circuit Details ---
R_SHUNT_OHMS = 100.0 # IMPORTANT: Measure this resistor for
best accuracy!
ADC_REF_VOLTAGE = 3.3 # Pico's reference voltage

# --- Sweep Parameters ---
NUM_STEPS = 256 # Number of voltage steps in the sweep
SETTLE_TIME_MS = 5 # Time for the RC filter and ADC to
stabilize
#
=====

# --- Pin Setup ---
pwm = machine.PWM(machine.Pin(PWM_PIN_NUM))
adc_v_applied = machine.ADC(machine.Pin(ADC_V_APPLIED_PIN_NUM))
adc_v_shunt = machine.ADC(machine.Pin(ADC_V_SHUNT_PIN_NUM))

# Set a PWM frequency. 100kHz is fast enough to be smoothed well.
pwm.freq(100_000)

# --- Data Storage ---
results = []

def u16_to_volts(u16_val):
    """Converts a 16-bit ADC reading to a voltage."""
    return u16_val * (ADC_REF_VOLTAGE / 65535)

def main():
    print("=" * 40)
    print("Diode I-V Curve Tracer")
    print(f"Sweeping 0V to {ADC_REF_VOLTAGE}V in {NUM_STEPS}
steps.")
    print("=" * 40)

    try:
        for i in range(NUM_STEPS + 1):
            # 1. Set the stimulus voltage by sweeping the duty
cycle
            duty = int(i * (65535 / NUM_STEPS))
            pwm.duty_u16(duty)

            # 2. Wait for the RC filter to settle
            time.sleep_ms(SETTLE_TIME_MS)

            # 3. Read the response
            # Average a few readings to reduce noise
            v_applied_u16 = sum(adc_v_applied.read_u16() for _ in
range(10)) // 10
            v_shunt_u16 = sum(adc_v_shunt.read_u16() for _ in
range(10)) // 10

            # 4. Convert readings to real-world units
            v_applied = u16_to_volts(v_applied_u16)

```

```

        v_shunt = u16_to_volts(v_shunt_u16)

        # Use Ohm's Law to find current
        i_diode = v_shunt / R_SHUNT_OHMS

        # Calculate the actual voltage across the diode
        v_diode = v_applied - v_shunt

        # Store the (V, I) pair
        results.append((v_diode, i_diode))

    # Optional: Print a progress bar
    progress = int((i / NUM_STEPS) * 30)
    print(f"[{'#' * progress}{'-' * (30 - progress)}]")
    Vd={v_diode:.2f}V, I={i_diode*1000:.2f}mA", end='\r')

    print("\n" + "="*40)
    print("Sweep Complete!")

    # --- Save Data as CSV ---
    print("Saving data to diode_curve.csv...")
    with open("diode_curve.csv", "w") as f:
        f.write("V_diode (V),I_diode (A)\n") # Write header
        for v, i in results:
            # Avoid negative voltage/current due to tiny
            # measurement errors at 0V
            if v < 0: v = 0
            if i < 0: i = 0
            f.write(f"{v},{i}\n")
        print("Save complete.")

except KeyboardInterrupt:
    print("\nSweep stopped by user.")
finally:
    # Turn off the PWM
    pwm.duty_u16(0)
    pwm.deinit()
    print("PWM de-initialized.")

main()

```

The code is pretty simple, and the comments should make it clear.

We save the data in a CSV file to graph on a computer that has a screen.

The graphing code is even simpler:

```

import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

CSV_FILENAME = "diode_curve.csv"

try:

```

```

df = pd.read_csv(CSV_FILENAME)
except FileNotFoundError:
    print(f"Error: {CSV_FILENAME} not found.")
    exit()

# Extract columns
v_diode = df["V_diode (V)"]
# Convert current to milliamps for a more readable plot
i_diode_mA = df["I_diode (A)"] * 1000

# --- Find the Forward Voltage (Vf) ---
# A common way to define Vf is the voltage at which 1mA of current
flows.
vf_current_threshold_mA = 1.0
vf = np.interp(vf_current_threshold_mA, i_diode_mA, v_diode)

# --- Create the Plot ---
plt.figure(figsize=(10, 7))
plt.plot(v_diode, i_diode_mA, marker='.', linestyle='--',
label='Diode I-V Curve')

# --- Add Annotations ---
if vf > 0.1: # Only annotate if a forward voltage was clearly found
    plt.axvline(x=vf, color='r', linestyle='--', label=f'Vf @
{vf_current_threshold_mA}mA = {vf:.3f} V')
    plt.axhline(y=vf_current_threshold_mA, color='g',
linestyle=':')

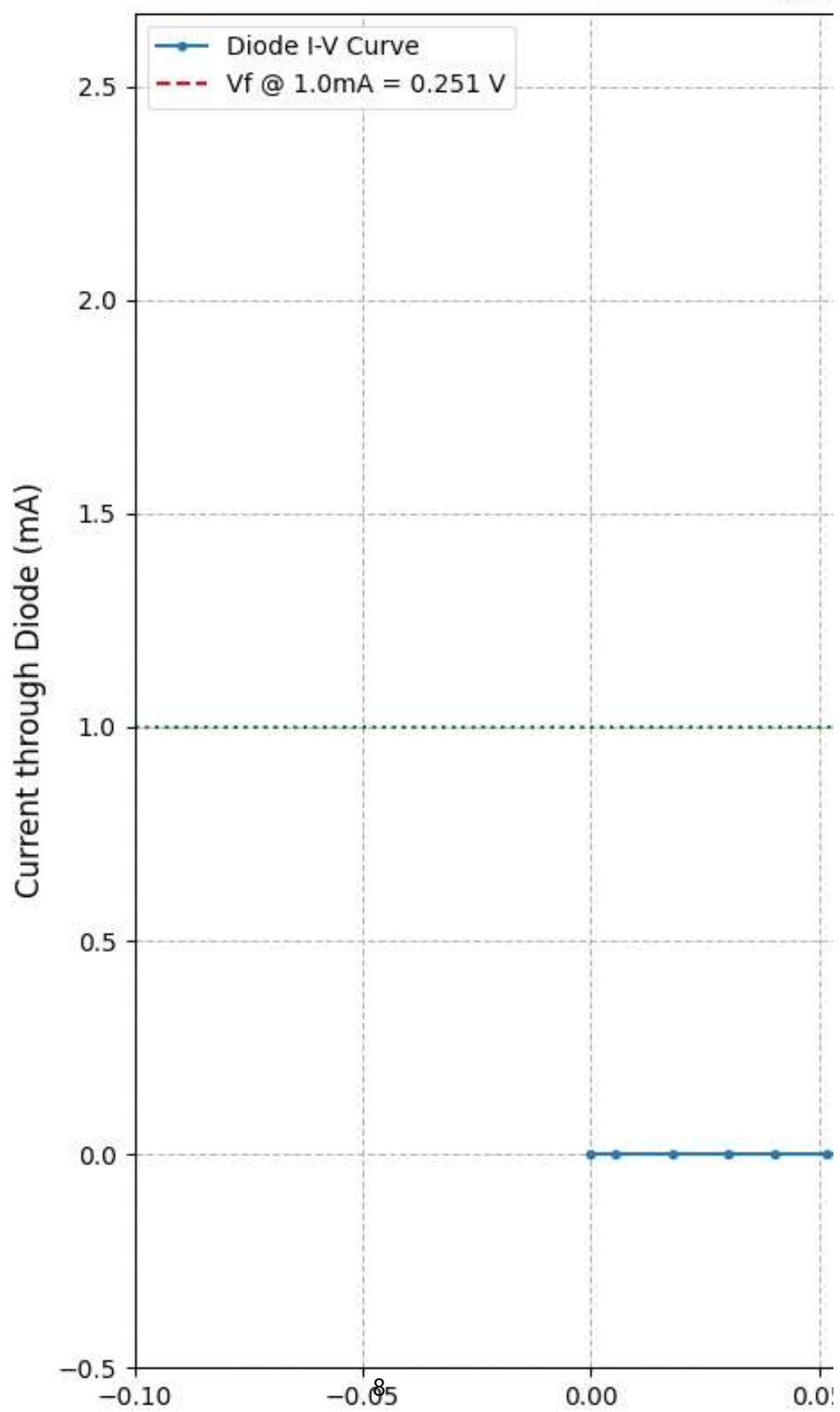
# --- Formatting ---
plt.title("Diode I-V Characteristic Curve", fontsize=16)
plt.xlabel("Voltage across Diode (V)", fontsize=12)
plt.ylabel("Current through Diode (mA)", fontsize=12)
plt.grid(True, which="both", ls="--")
plt.legend()
plt.ylim(bottom=-0.5) # Start y-axis slightly below zero
plt.xlim(left=-0.1)   # Start x-axis slightly below zero
plt.show()

```

The Test

Before we build our own diodes, let's test our code on some commercially available modern diodes. One diode commonly used in crystal radios is the 1N34A germanium diode:
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Dio



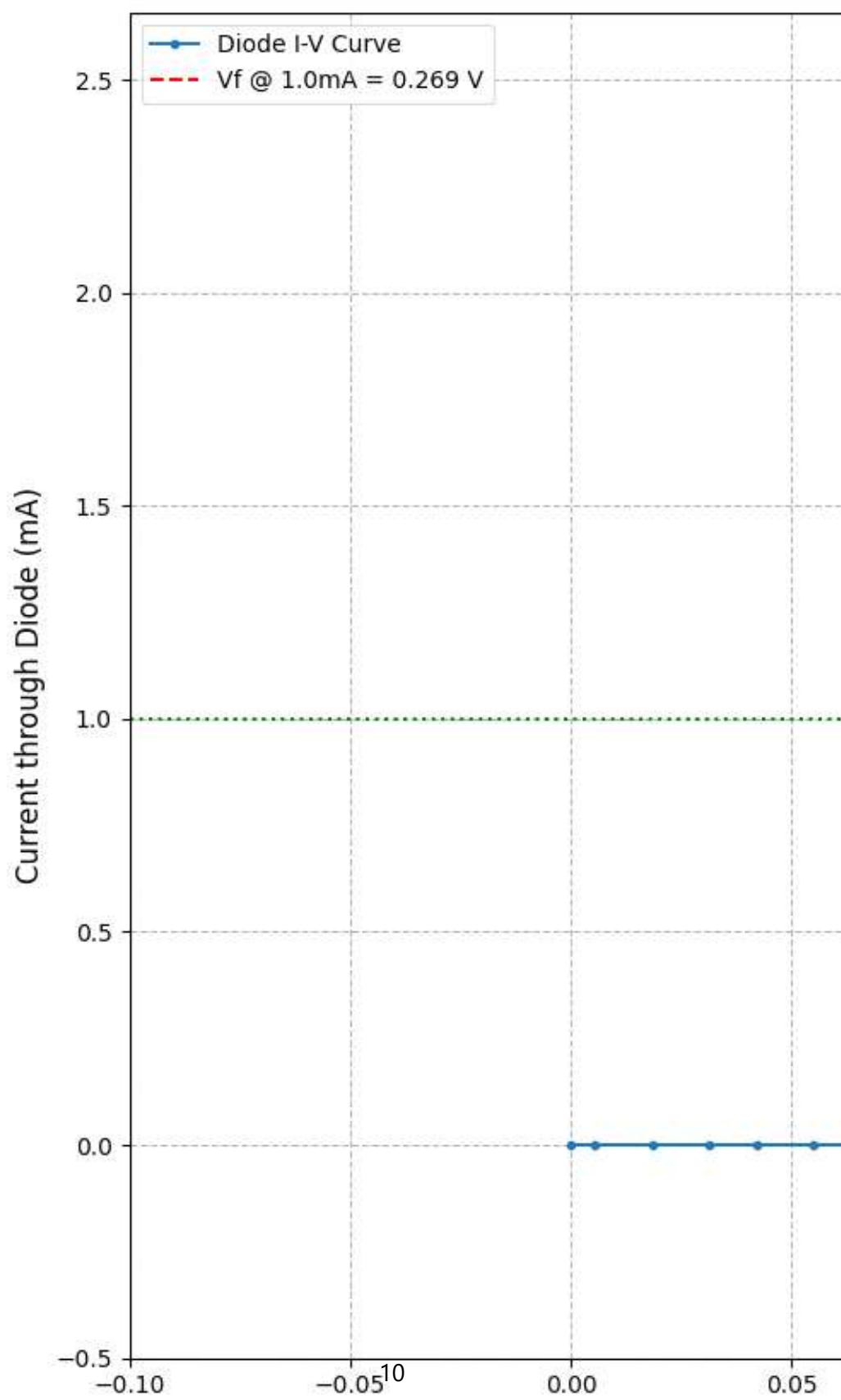
Results for a 1N34A Germanium Diode

We can see that it begins to conduct at about 140 millivolts, and we get a full milliampere of current at 251 millivolts. That comes to a hair over a quarter milliwatt. With a sensitive earphone, that is plenty of power for a nice, loud signal.

Another favourite for crystal radios is a Schottky diode, such as the BAT46:

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Dio



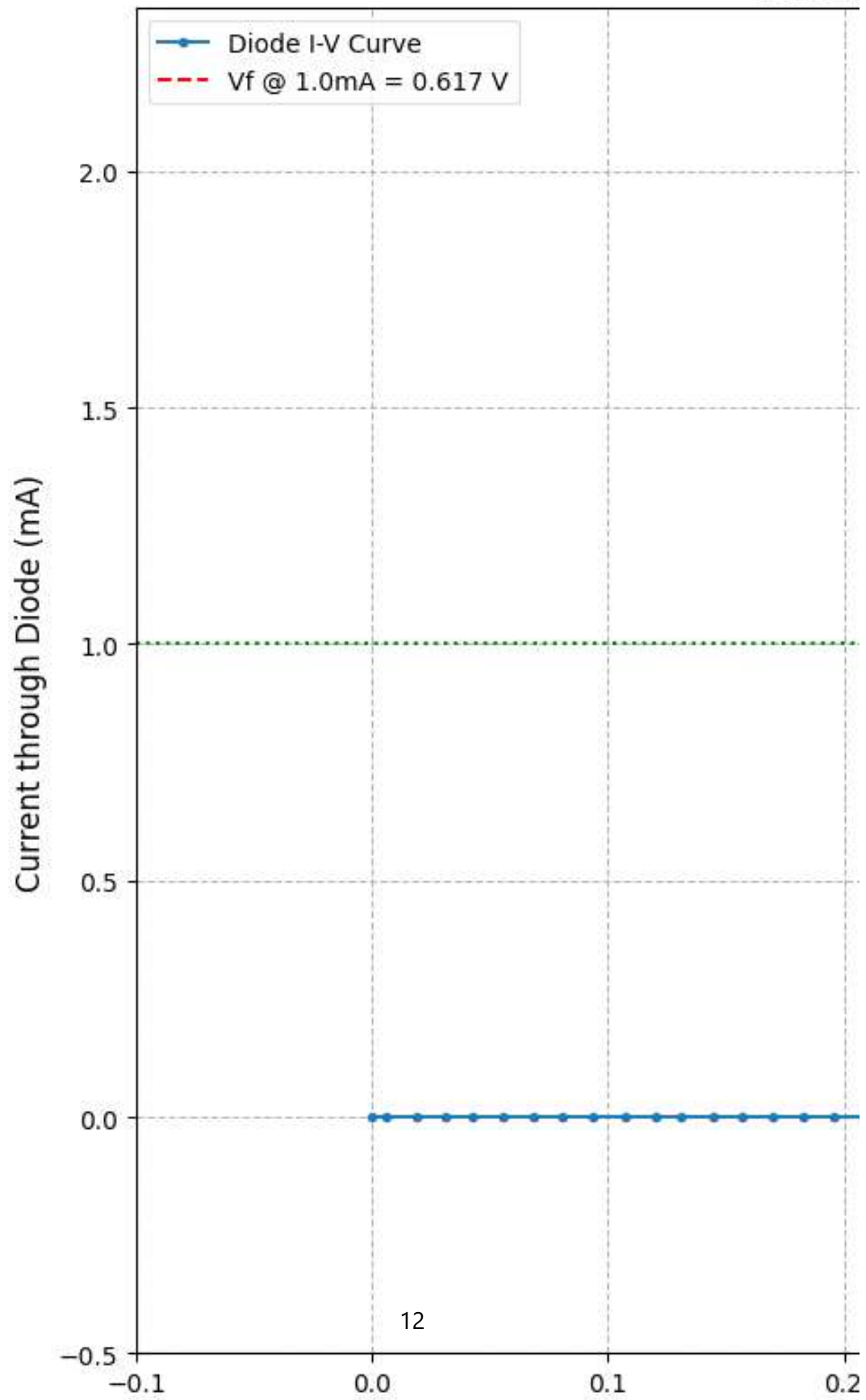
BAT46 Schottky Diode I-V Curve

It is just a bit less sensitive than the 1N34A.

If you have a strong AM station nearby, you can use a very common, cheap silicon diode, such as the 1N914:

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Diode



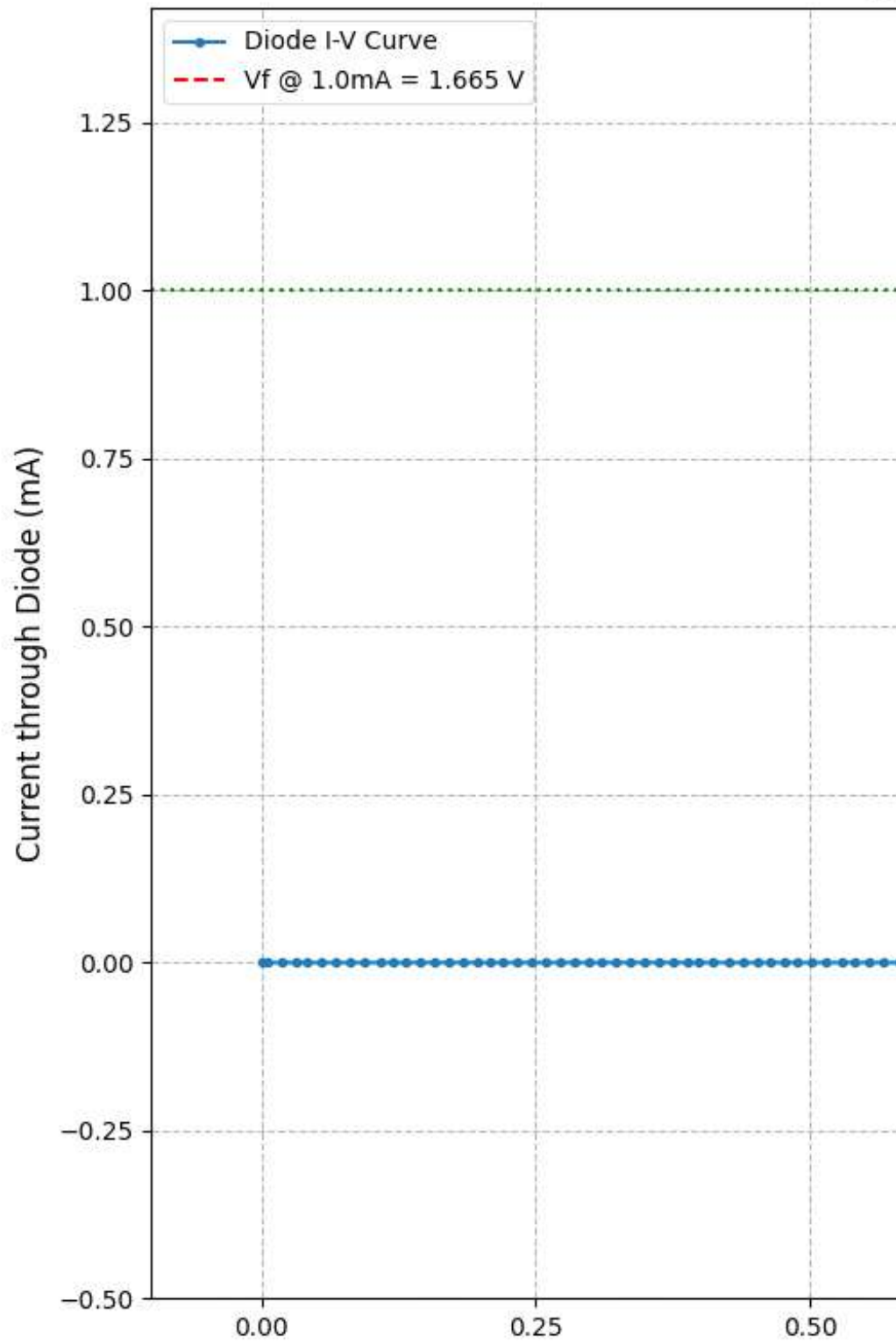
1N914 Silicon Diode I-V Curve

Notice that it needs over half a milliwatt, so you might want a long antenna.

Since I have a very long antenna, I can even light up a red LED in my crystal radio:

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D



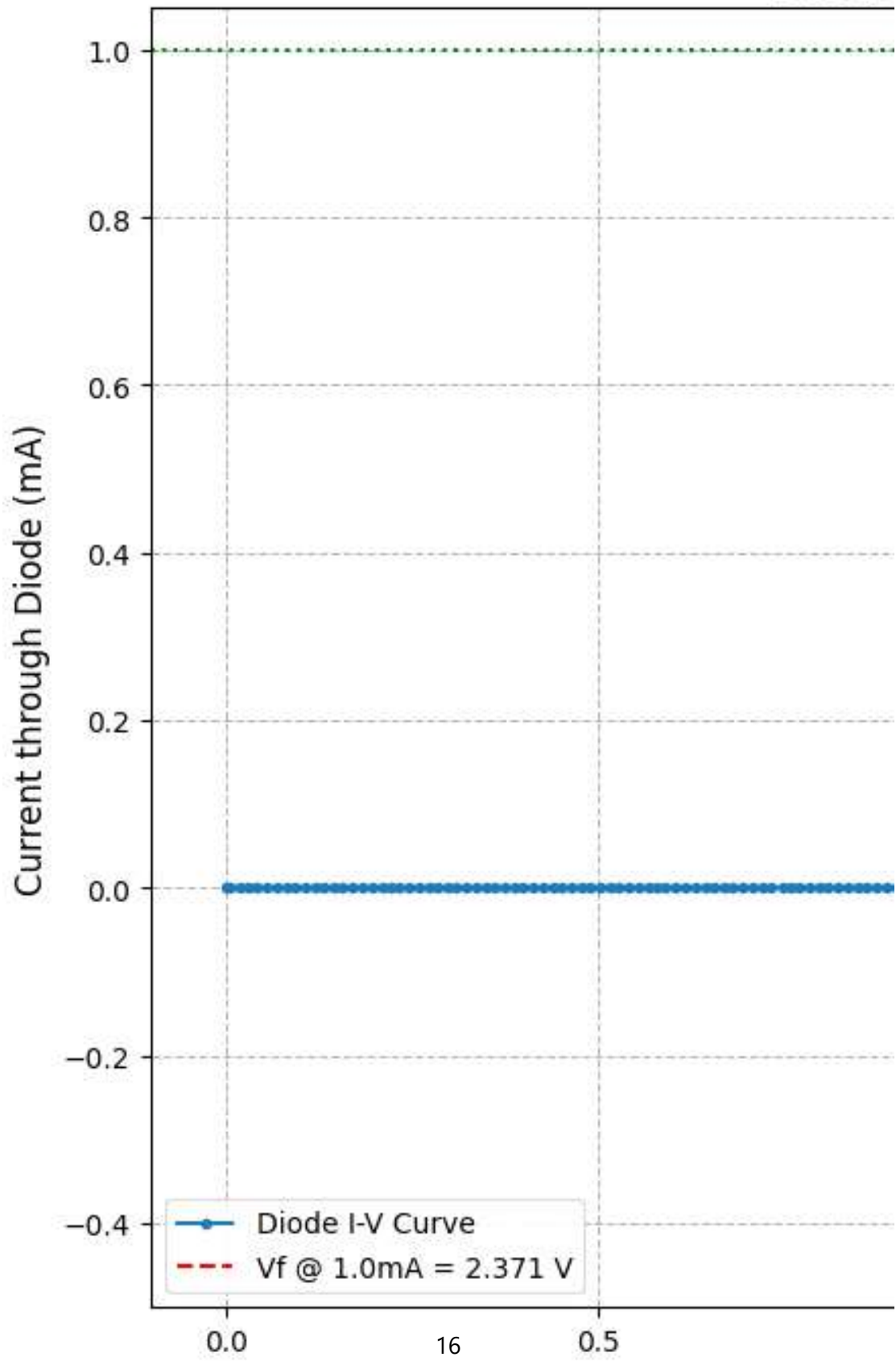
Red LED I-V Curve

Over one and a half milliwatts of power.

The colour of an LED relates to the voltage at which it turns on. The bluer the LED, the higher the voltage. A green LED barely turns on in my radio.

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Diode



Green LED I-V Curve

Building Our Own

Now it is time to build our own diodes. Our first one will be a cuprous oxide diode. Copper has two types of oxide. One is black cupric oxide, CuO . The other is the salmon-colored cuprous oxide Cu_2O . We want to make that second kind, the one that is a semiconductor.

We start by heating up a small sheet of copper.
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We can see the oxide layers start to form. But we want a nice thick layer, so we continue heating.

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After a while, the pink oxide has turned black. But that's OK, because we can remove the black cupric oxide, and underneath is the pink cuprous oxide.

Using a pair of long-nosed pliers, we take it off the burner and drop it into water.

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Most of the black oxide flakes off as the water boils around the copper.

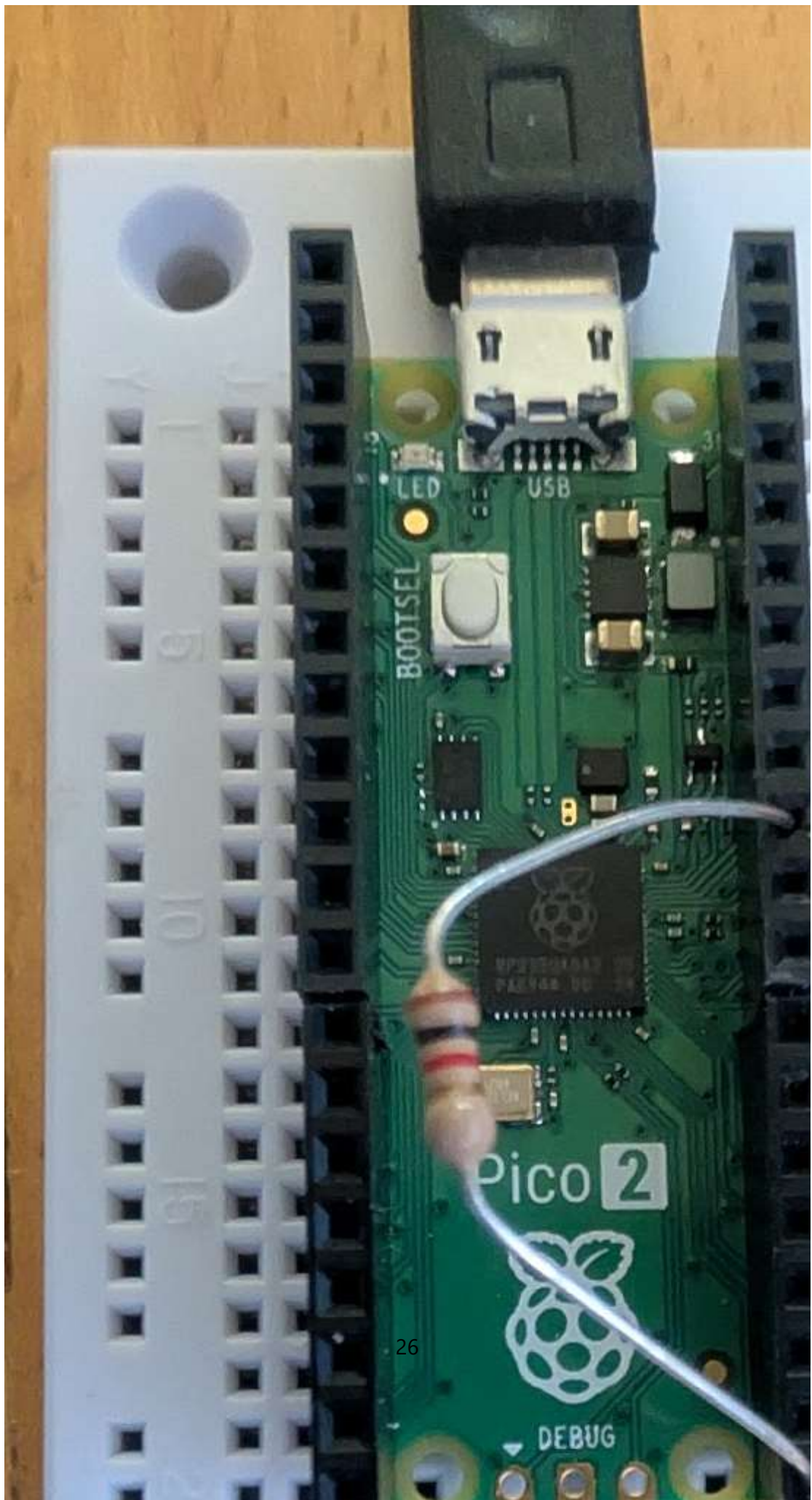
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I made several diodes this way. Some flaked clean but did not work well, and some had lots of black oxide left, but worked out.

Here is the test setup:

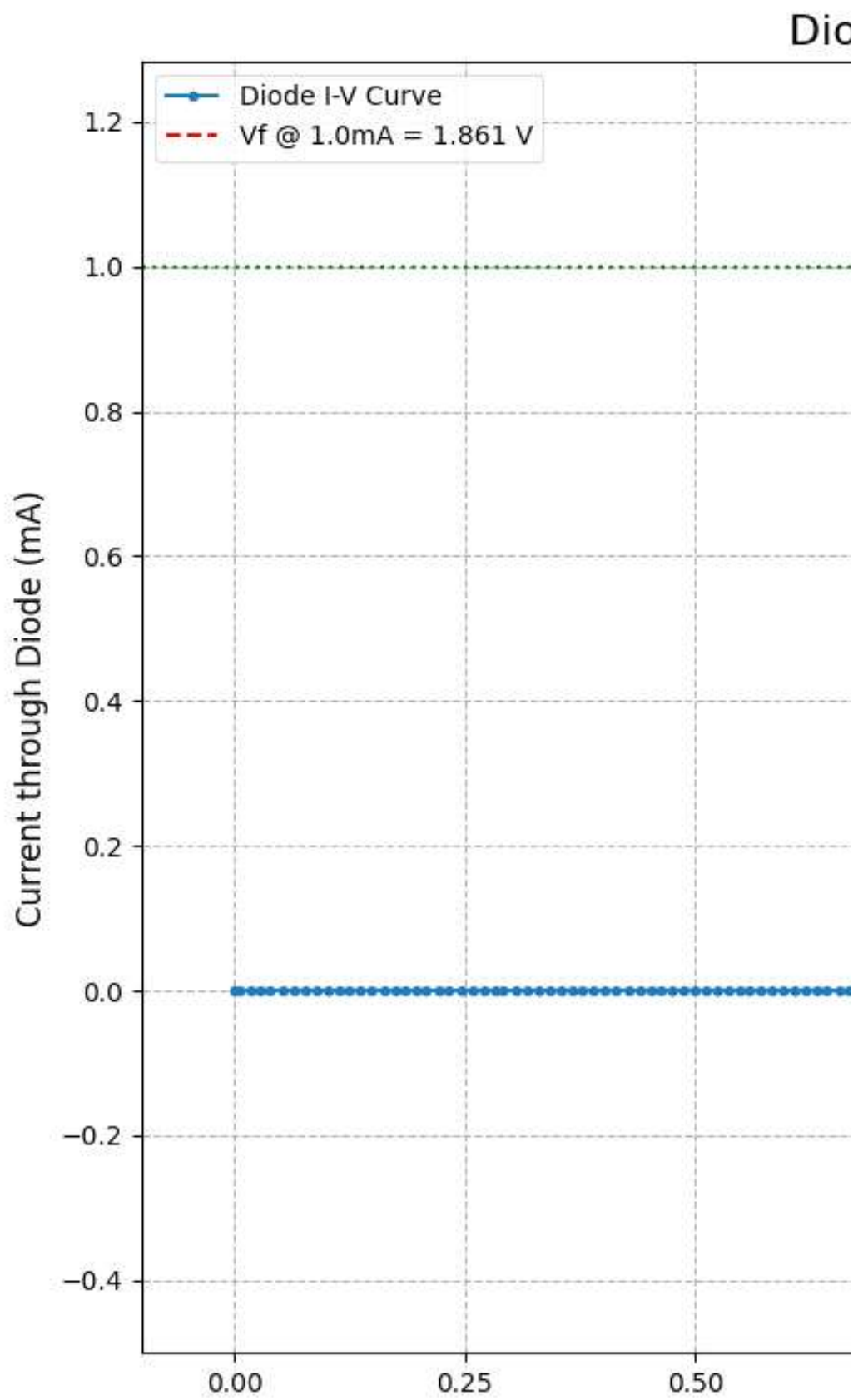
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The “cat’s whisker” is a pencil lead. Graphite on cuprous oxide worked, while harder metals scratched through the oxide and shorted out. Lead or tin might work, but I did not try those.

Here is how the diode performed:

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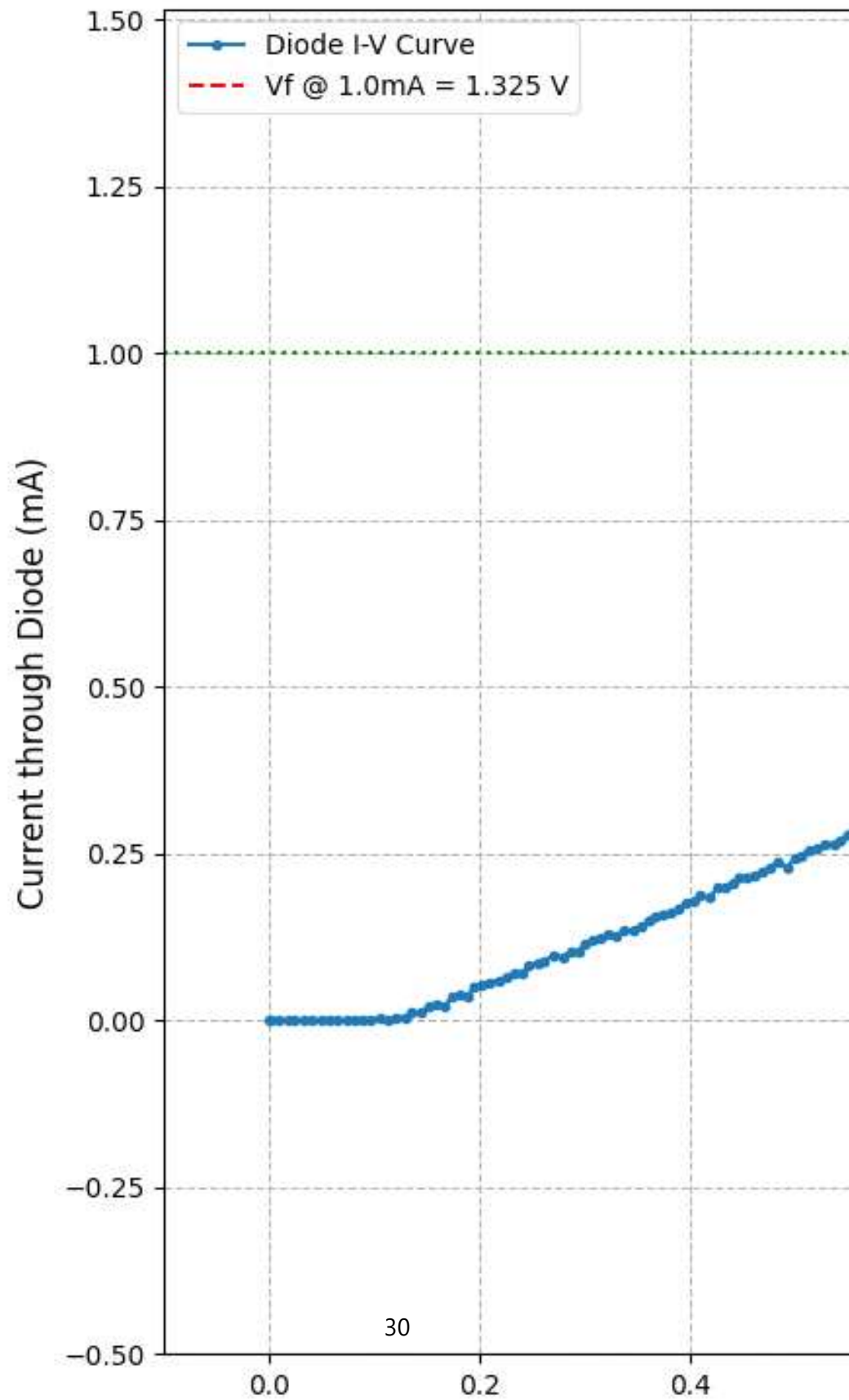


Homemade Cuprous Oxide / Graphite Diode

The thin oxide layer reacts a bit when the voltage gets higher, but this diode pulled in four AM radio stations.

Moving the pencil lead around can improve things a bit. This location barely made the graph curve, but brought the sensitivity up a little.
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Dioc



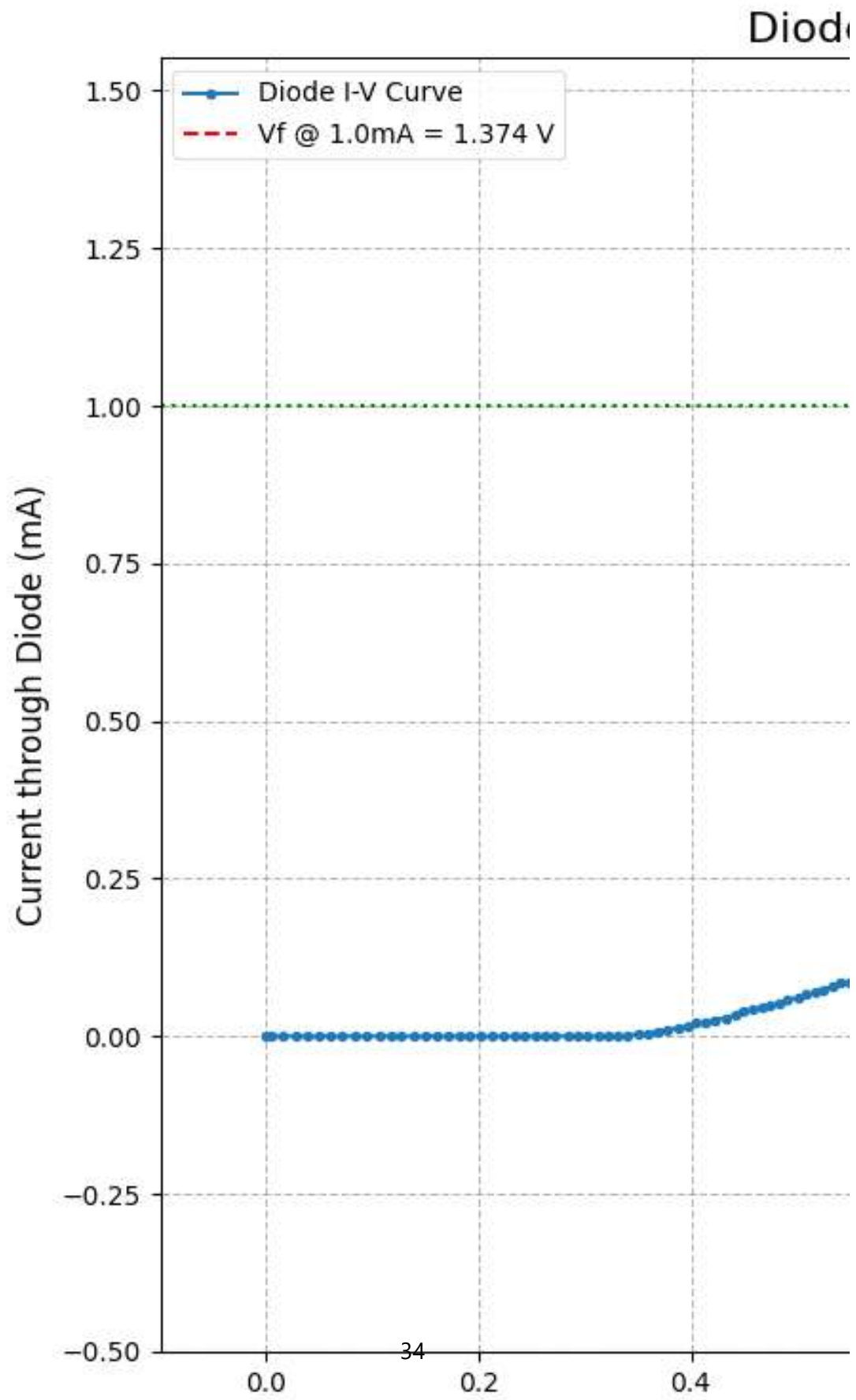
Homemade Cuprous Oxide / Graphite Diode

Next, I made a blue steel diode, the classic “Foxhole Radio” diode used by soldiers in World War II.

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You heat up a carbon steel blade and use the same pencil lead.
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Blue Steel / Graphite Diode I-V Curve

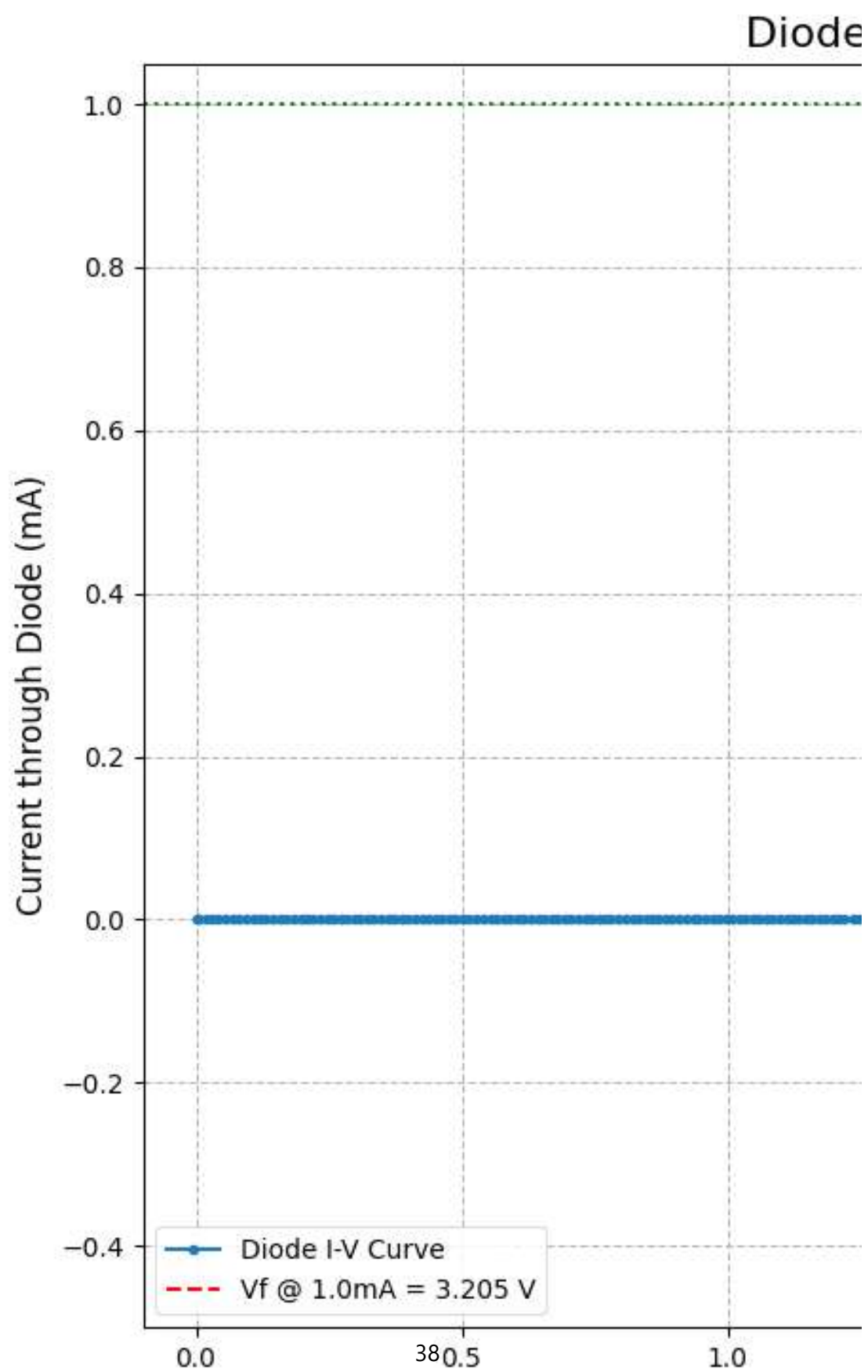
Not bad. Not nearly as sensitive as the 1N34A, but it brought in those same 4 stations.

I have a beautiful chunk of carborundum, crystals of silicon carbide, SiC.

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It gets used for some high-power semiconductor devices and even some early blue LEDs. Unlike the first two homemade diodes, it needs some force to make contact, so I placed a tiny bit of it between two brass sheets and held them together with a strong binder clip. [Press enter or click to view image in full size](#)



Homemade Carborundum Diode I-V Curve

You can see why it was used for blue LEDs. It barely conducts at the maximum voltage our RP2350 can provide. But we can make it more sensitive by cheating. If we use a couple of batteries to add 3 volts, our radio signal can now reach the diode's exponential portion of the curve.

Other minerals, oxides, and sulfides can be used to make homemade diodes. Some of the common ones are iron pyrite, galena, and zincite.

Now that you have a \$6 diode curve tracer, you can experiment with making your own.